



ত্রিশাল, ময়মনসিংহ

Project Report On

“SMART FARMING SYSTEM”

Course Code: CSE-100

Course Name: Software Development Project With C/Python

Submitted by:

Md. Hussain Anjum Ratul

Roll: 22102041

Reg: 10847

Biplob Chandra Das

Roll: 22102027

Reg: 10833

Supervised by:

Dr. Md. Selim Al Mamun

Professor

Dept. of Computer Science and Engineering

Jatiya Kabi Kazi Nazrul Islam University

BOARD OF EXAMINERS

1. Name of the Supervisor

Chairman

Dr. Md. Selim Al Mamun

Professor

Dept. of Computer Science and Engineering

Jatiya Kabi Kazi Nazrul Islam University

Signature

2. Name of the Internal Member

Member

Signature

3. Name of the Internal Member

Member

Signature

4. Name of the External Member

Member

Signature

As a supervisor, I authorize that the work is done under my guidance. I have approved the submission of that report to go for an evaluation.

Date: 15.01.2024

.....
Signature
Dr. Md. Selim Al Mamun
Professor

CANDIDATE'S DECLARATION AND SUPERVISOR'S APPROVAL

As candidates, We declare that this project is our work and has not been submitted in any other form for another degree or diploma at any university or other institute of tertiary education. Information derived from the published and unpublished work of others has been acknowledged in the text and a list of references is given.

Date: 15.01.2024

.....
Signature
Md. Hussain Anjum Ratul

Date: 15.01.2024

.....
Signature
Biplob Chandra Das

Acknowledgment

A project is a golden opportunity for learning and self-development. Many people deserve our cordial thanks for their help in completing this project. First and foremost, we would like to thank our honorable Supervisor Dr. Md. Selim Al Mamun, Professor, Department of CSE, JKKNIU who is so dedicated to helping us in our times of need. We are honored to make this project under his guidance that provides us with his valuable time and knowledge, motivating thought, and encouragement.

We wish to express thanks to DR. MD. SUJAN ALI, Head, Department of Computer Science and Engineering, Jatiya Kabi Kazi Nazrul Islam University for providing us with lab opportunities and lab materials related to our project work.

Finally, we express our thanks to the Department of Computer Science and Engineering for allowing us to study here and for supporting us greatly.

Abstract

The “SMART FARMING SYSTEM” is an application developed using Tkinter and Python, focusing on providing agricultural information and crop distribution details for the Mymensingh district in Bangladesh. The application features a secure login system for admins, allowing users to input district and sub-district preferences. The primary functionalities include searching for agricultural information and accessing details about crop distribution in specific sub-districts. Admins can also update and modify information through an interactive interface. The project incorporates a visually appealing design with background images and buttons for an enhanced user experience.

Table of Contents

Acknowledgement.....	i
Abstract.....	ii
1. Introduction	4
1.1 Introduction	4
1.2 Objectives	4
1.3 Motivation of the Project	5
2. Background Study	6
2.1 Background	6
2.2 Functional Requirements	6
3. Methodology	7
3.1 Application Flow	7
3.2 Algorithm	8
3.3 Modules	9
4. Design and Implementation	10
4.1 User Interface	10
4.2 Development Tools and Environments	13
5. Conclusion	14
5.1 Conclusion	14
5.2 Limitations of this Project	14
5.3 Future Work	14
References.....	15

Chapter-1

INTRODUCTION

1.1 Introduction

The “SMART FARMING SYSTEM” is a Tkinter-based application designed to provide agricultural information and crop distribution details for the Mymensingh district in Bangladesh. The application features a main window with a fixed size and a background image. It incorporates a login frame for admins to access additional functionalities. The search frame enables users to select a district and sub-district for obtaining agricultural information. Upon successful selection, the application displays relevant details and crop distribution for the chosen sub-district in separate windows. Admins have the capability to update information using a file dialog. A feedback button opens a dedicated window for users to provide feedback, stored in a text file. Overall, it provides a user-friendly interface for accessing agricultural information and facilitates interactive features for both users and admins.

This system aims to streamline data access and management, empowering administrators with the necessary insights for informed decision-making in agriculture.

1.2 Objectives

The major objectives of this application project are:

- **User Interface Design:** Create an aesthetically pleasing and intuitive graphical user interface (GUI) with background images, labels, buttons, and appropriate font styles
- **District and Sub-district Selection:** Allow users to select a specific district and sub-district from predefined options through ComboBox widgets
- **Search Functionality:** Implement a search mechanism to validate and confirm the selected district and sub-district. Provide appropriate feedback to the user
- **Admin Authentication:** The application requires authentication with an admin username and password
- **Display Results:** Display a confirmation message indicating the selected sub-district and offer options to explore further information or crop distribution details
- **Information Retrieval:** Retrieve and display agricultural information for the selected sub-district from stored text files. Create a user-friendly interface for information presentation

- **Customizable Data:** Store information and crop distribution data in text files, facilitating easy customization and updates for each sub-district
- **Update Information:** The application allows the admin to update agricultural information by selecting a text file through a file dialog
- **Feedback and Communication:** Users can provide feedback and communicate with the admin through a separate window for feedback
- **Error Handling:** Implement error messages to notify users of invalid district or sub-district selections, guiding them to provide accurate information. To add game restart system

1.3 Motivation of the Project

The motivation behind the "SMART FARMING SYSTEM" project stems from a genuine commitment to enhance agricultural knowledge and accessibility in the Mymensingh district of Bangladesh. The agricultural sector is a vital component of the economy, and empowering farmers and stakeholders with information can significantly contribute to sustainable farming practices and improved yields.

The "SMART FARMING SYSTEM" project is motivated by the aspiration to leverage technology for the betterment of agricultural communities in the Mymensingh district, fostering empowerment, sustainability, and community development.

Chapter-2

BACKGROUND STUDY

2.1 Background

The “SMART FARMING SYSTEM” project emerges from the pressing need to leverage technology for efficient agricultural practices in the Mymensingh district of Bangladesh. Recognizing the pivotal role of agriculture in the region’s economy, the project aims to empower farmers and stakeholders with a comprehensive information system.

The background is rooted in the vision of creating a user-friendly platform that provides not only agricultural information and crop distribution data but also facilitates communication between admins and users.

By incorporating features like admin authentication, localized content, and file-based information updates, the project seeks to bridge the technological gap and enhance decision-making processes in the agricultural sector.

The background is underscored by a commitment to smart farming, sustainability, and community engagement, aligning with the broader goal of advancing agricultural practices for a more prosperous and resilient future in the Mymensingh district.

2.2 Functional Requirements

The functional requirement for making an application such as "SMART FARMING SYSTEM" requires the use of many software technologies. Which include a programming language, in this case Python (Python is a great choice for application development due to its simple syntax, powerful libraries, and ease of integration with other languages and tools. Python is also highly extensible, meaning that it can be used to write custom scripts and modules for applications. Additionally, Python has a large and active community of developers, which makes it easy to find help and resources when developing applications. Finally, Python is open source, meaning that it is free to use and modify), a code editor (e.g. Vs Code), graphic design softwares (e.g. Photoshop)

Chapter-3

METHODOLOGY

3.1 Application Flow

The application flow of the “SMART FARMING SYSTEM” project begins with the user launching the Tkinter-based interface. The main window features a visually appealing background image and prompts the admin for authentication, allowing only authorized admin to access certain functionalities. The user is presented with a search interface where they can choose a district and sub-district (Upazila) for agricultural information retrieval. The system validates the inputs and displays relevant information along with buttons to explore detailed agricultural data and crop distribution results for the selected sub-district. Admins can update information by selecting a file through a file dialog. The application also offers a user feedback option, allowing users to submit their thoughts and suggestions. The project emphasizes a localized Bengali user interface, contributing to a more accessible and user-friendly experience. The flow concludes with the option to close the main window or provide feedback, reflecting a systematic and intuitive design.

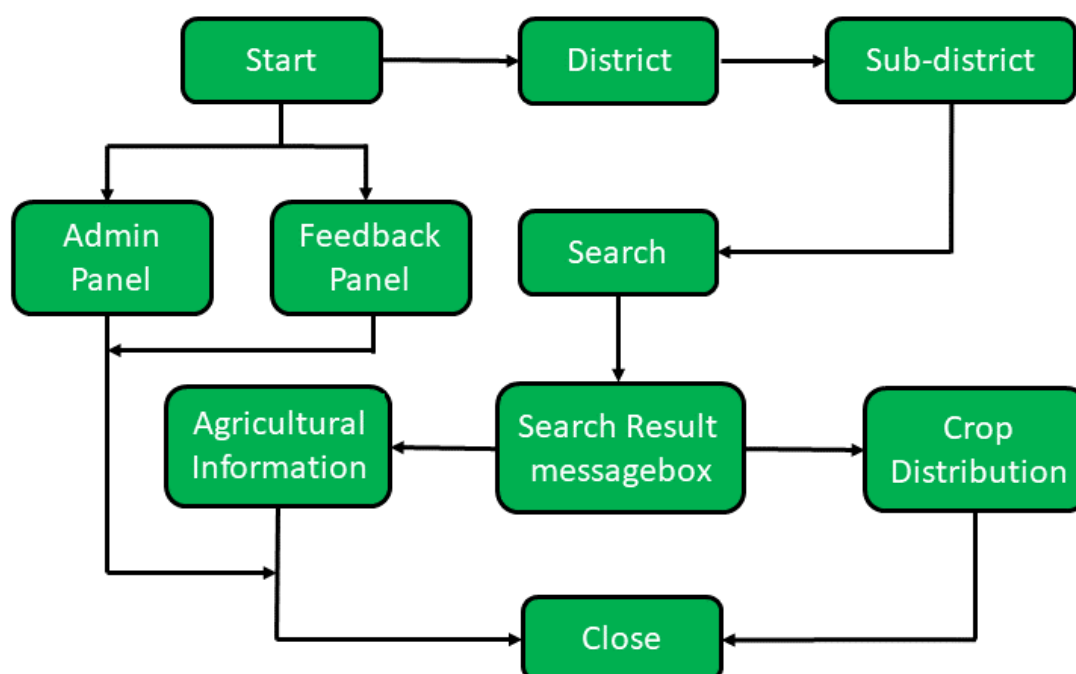


Fig 3.1: Application Diagram

3.2 Algorithm

Step 1: Start

Step 2: Import necessary libraries: tkinter, PIL (Pillow), messagebox, ttk, filedialog.

Step 3: Create Main Application Class. Define the SearchApp class inheriting from Tkinter's Tk class.

Step 4: In the constructor (__init__), set the title of the main window as "SMART FARMING SYSTEM". Set the geometry of the main window to "1920x1080". Load and display a background image on the main window.

Step 5: Create a frame (login_frame) to hold login elements. Add labels for username and password, entry fields, and a login button. Use the authenticate method for login functionality.

Step 6: Create a frame (search_frame) for search-related elements. Add labels for district and sub-district, Comboboxes, and a search button. Use the search method for search functionality.

Step 7: Implement the feedback method to create a new window for user feedback. Include a text widget for user input and buttons for submission and closing.

Step 8: Implement the authenticate method for admin authentication. Check entered credentials against predefined valid values. If authenticated, enable an "Update Information" button for the admin.

Step 9: Implement the update_file method to allow the admin to update text files. Use a file dialog to select a file, open it for updating, and create a new window for user input. Include a button for submitting the updated content and closing the window.

Step 10: Implement the search method for processing search requests. Check the selected district and sub-district against valid values. Display the selected sub-district and offer buttons for "Info" and "Result."

Step 11: Implement methods (show_info and show_result) for displaying information and results. Retrieve file paths from dictionaries based on the selected sub-district. Read file content and display it in a new window with appropriate formatting. Include buttons for closing the information and result windows.

Step 12: Add a button to close the main window.

Step 13: Create an instance of the Tk class, instantiate the SearchApp class, and start the main event loop.

Step 14: End

3.3 Modules

A Python module is a file containing Python definitions and statements. A module can define functions, classes, and variables. A module can also include runnable code. Grouping related code into a module makes the code easier to understand and use. The “SMART FARMING SYSTEM” project incorporates several modules to facilitate its diverse functionalities. These modules can be categorized as follows:

- **GUI Design Module:** Handles the graphical user interface (GUI) design using the Tkinter library, incorporating labels, buttons, entry widgets, comboboxes, and background images.
- **User Authentication Module:** Responsible for authenticating admins by validating their login credentials.
- **Search Module:** Manages the search functionality, enabling users to select a district and sub-district for retrieving agricultural information.
- **Display Information Module:** Handles the display of agricultural information based on the selected sub-district, including details on crops and farming practices.
- **Update Information Module:** Allows admins to update agricultural information by selecting a text file through a file dialog and provides a user-friendly interface for content modification.
- **Feedback Module:** Manages the feedback functionality, enabling users to submit their thoughts, suggestions, and comments.
- **File Handling Module:** Manages file operations, such as reading and writing information to text files, for storing agricultural details, user feedback, and admins updates.
- **Localization Module:** Incorporates localization features by providing text labels and instructions in the Bengali language for a more user-friendly experience.
- **Communication Module:** Facilitates communication between users and administrators, allowing for the submission of feedback and suggestions.
- **Close Window Module:** Handles the functionality to close the main window and additional windows created for displaying information, results, feedback, and updates.

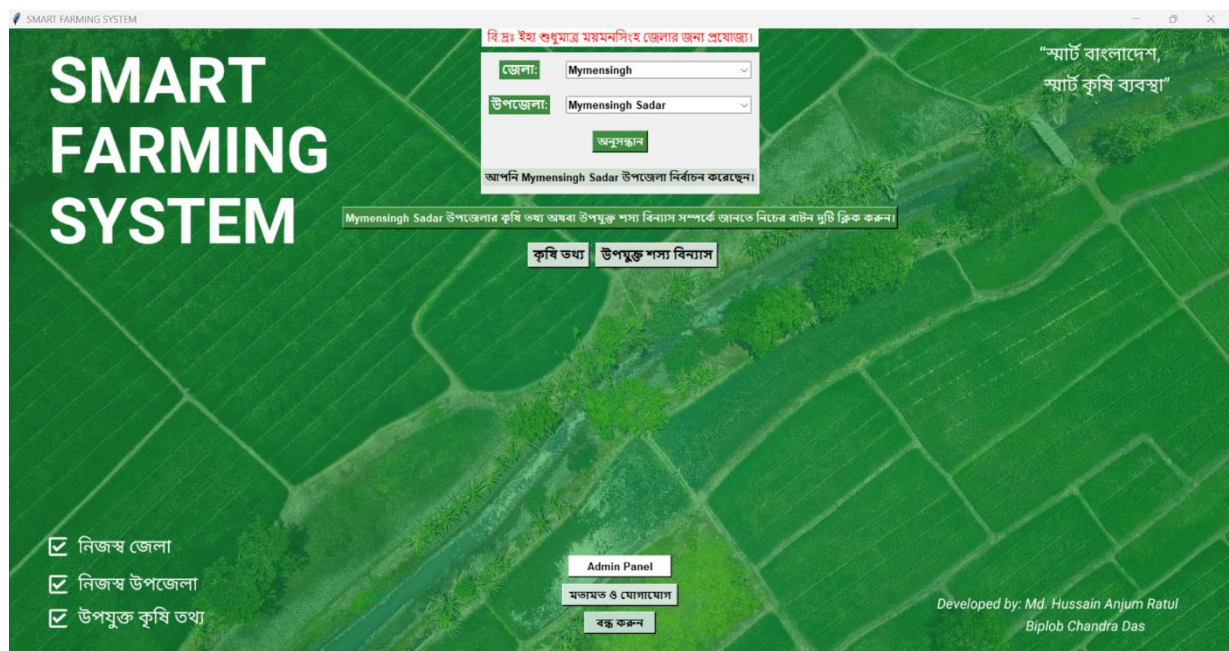
These modules work collaboratively to create a cohesive and functional "SMART FARMING SYSTEM," addressing various aspects such as information retrieval, data updating, user interaction, and system control.

Chapter-4

DESIGN AND IMPLEMENTATION

4.1 User Interface

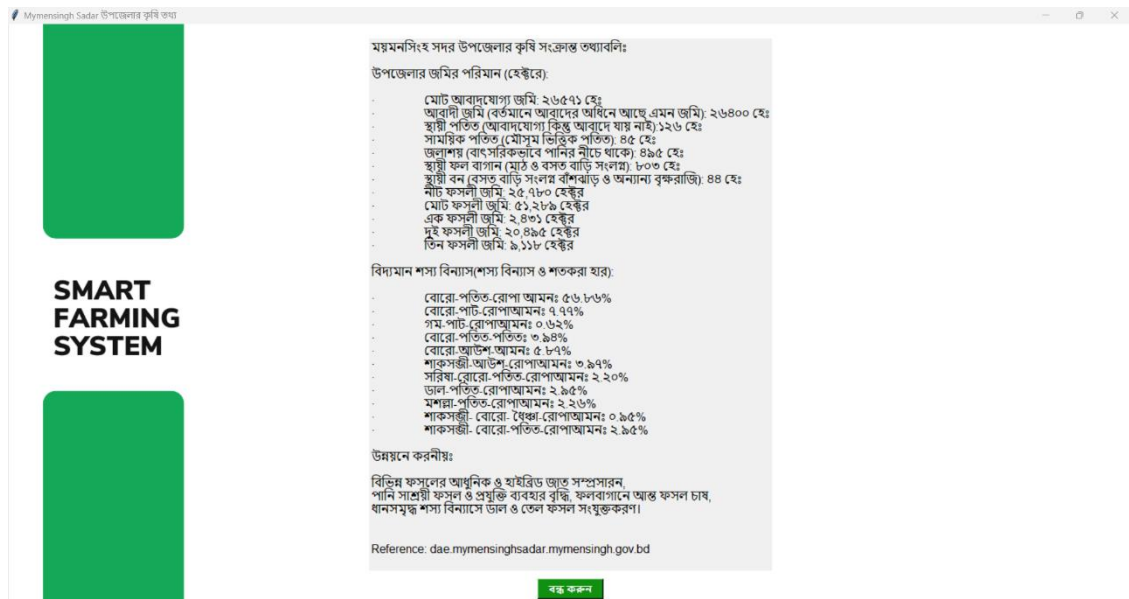
Home Page: The homepage has 2 panels. One is the user panel and the other is the admin panel.



User Panel: At this section user will get a confirmation message by selecting his own district and upazila. Along with this two buttons will pop up. These are: “কৃষি তথ্য” and “উপযুক্ত শস্য বিন্যাস”.



Agricultural Information window: By clicking on the “কৃষি তথ্য” button, the user can know about the agricultural information of his upazila in this window.



Mymensingh Sadar উপজেলার কৃষি তথ্য

ময়মনসিংহ সদর উপজেলার কৃষি সংক্রান্ত তথ্যাবলি:

উপজেলার জমির পরিমান (হেক্টরে):

- মোট আবাসযোগ্য জমি: ২৬৫৭১ হেক্টর
- অবৈধ জমি (বর্তমানে আবাদের অধিনে আছে এমন জমি): ২৬৪০০ হেক্টর
- স্থায়ী পতিত (আবাদযোগ্য কিন্তু আবাদে যায় নাই) ১২৬ হেক্টর
- সাময়িক পতিত (মৌসুম ভিত্তিক পতিত): ৪৫ হেক্টর
- জলাশয় বাৎসরিকভাবে পানির নিচে থাকে: ৪৯৫ হেক্টর
- স্থায়ী ফল বাগান (মুঠ ও বসত বাড়ি সংলগ্ন): ৮০৩ হেক্টর
- স্থায়ী বন (বসত বাড়ি সংলগ্ন বাঁশঝাড় ও অন্যান্য বৃক্ষরাজি): ৪৪ হেক্টর
- মোট ফসলী জমি: ২৫,৭৮০ হেক্টর
- মোট ফসলী জমি: ৫১,২৮৯ হেক্টর
- এক ফসলী জমি: ২,৪৩১ হেক্টর
- দুই ফসলী জমি: ২০,৪৯৫ হেক্টর
- তিন ফসলী জমি: ৯,১১৮ হেক্টর

বিদ্যমান শস্য বিন্যাস(শস্য বিন্যাস ও শতকরা হার):

- বোরো-পতিত-রোপা আমন: ৫৬.৮৬%
- বোরো-পাট-রোপা আমন: ৭.৭৭%
- গম-পাট-রোপা আমন: ০.৬২%
- বোরো-পতিত-পতিত: ৩.৯৩%
- বোরো-আউশ-আমন: ৫.৮৭%
- শাকসব্জী-আউশ-রোপা আমন: ৩.৯৭%
- সরিষা-বোরো-পতিত-রোপা আমন: ২.২০%
- ডাল-পতিত-রোপা আমন: ২.৯৫%
- মশর-পতিত-রোপা আমন: ২.২৬%
- শাকসব্জী-বোরো-বৈধ-রোপা আমন: ০.৯৫%
- শাকসব্জী-বোরো-পতিত-রোপা আমন: ২.৯৫%

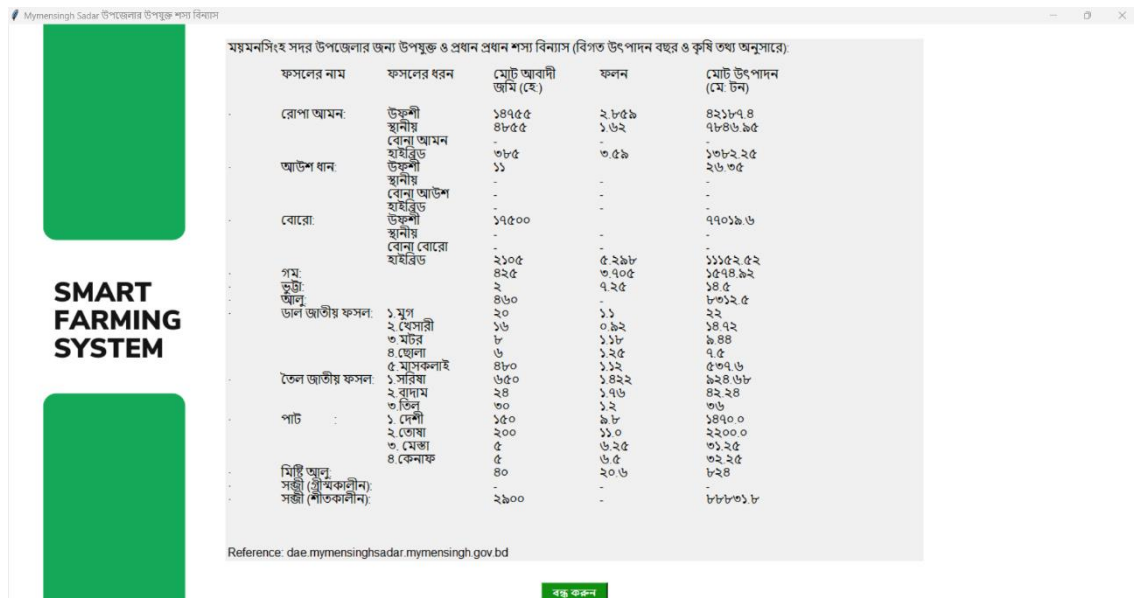
উন্নয়নে করণীয়:

বিভিন্ন ফসলের আধুনিক ও হাইব্রিড জাত সম্প্রসারণ,
পানি সাশ্রয়ী ফসল ও প্রযুক্তি ব্যবহার বৃদ্ধি, ফলবাগানে আশ্রয় ফসল চাষ,
ধানসমৃদ্ধ শস্য বিন্যাসে ডাল ও তেল ফসল সংযুক্তকরণ।

Reference: dae.mymensinghsadar.mymensingh.gov.bd

বন্ধ করুন

Suitable Crop Distribution window: By clicking on the “উপযুক্ত শস্য বিন্যাস” button, the user can know about the suitable crop distribution of his upazila in this window.



Mymensingh Sadar উপজেলার উপযুক্ত শস্য বিন্যাস

ময়মনসিংহ সদর উপজেলার জন্য উপযুক্ত ও প্রধান প্রধান শস্য বিন্যাস (বিগত উৎপাদন বছর ও কৃষি তথ্য অনুসারে):

ফসলের নাম	ফসলের ধরন	মোট আবাসী জমি (হেক্টর)	ফলন	মোট উৎপাদন (মে টন)
রোপা আমন:	উষ্ণ	১৪৭৫৫	২.৮৫৯	৪২১৮৭.৪
	স্থানীয়	৪৮৫৫	১.৬২	৭৮৪৬.৯৫
	বোনা আমন	-	-	-
আউশ ধান:	হাইব্রিড	৩৮৫	৩.৫৯	১৩৮২.২৫
	স্থানীয়	১১	-	২৬.৩৫
	বোনা আউশ	-	-	-
বোরো:	হাইব্রিড	-	-	-
	উষ্ণ	১৭৫০০	-	৭৭০১৯.৬
	স্থানীয়	-	-	-
	বোনা বোরো	-	-	-
গম:	হাইব্রিড	২১০৫	৫.২৯৮	১১৫২.৫২
ভুট্টা:	হাইব্রিড	৪২৫	৩.৭০৫	১৫৭৪.৯২
আলু:	হাইব্রিড	২	৭.২৫	১৪.৫
ডাল জাতীয় ফসল:	১ মুগ	৪৬০	-	৮৩১২.৫
	২ বেসরাসী	২০	১.১	২২
	৩ মটর	১৬	১.১৮	১৪.৭২
	৪ ছোলা	৬	১.২৫	৭.৫
	৫ মাসকলাই	৪৮০	১.১২	৫৩৭.৬
তেল জাতীয় ফসল:	১ সরিষা	৬৫০	১.৪২২	৯২৪.৬৮
	২ রানাম	২৪	১.৭৬	৪২.২৪
	৩ তিল	৩০	১.২	৩৬
পাট:	১ দেশী	১৫০	৯.৮	১৪৭০.০
	২ তোষা	২০০	১১.০	২২০০.০
	৩ মেজা	৫	৬.২৫	৩১.২৫
	৪ ফেনাফ	৫	৬.২৫	৩১.২৫
মিষ্টি আলু:	হাইব্রিড	৪০	২০.৬	৮২৪
সব্জী (শীতকালীন):	হাইব্রিড	২৯০০	-	-
সব্জী (শীতকালীন):	হাইব্রিড	২৯০০	-	৮৮৮৩১.৮

Reference: dae.mymensinghsadar.mymensingh.gov.bd

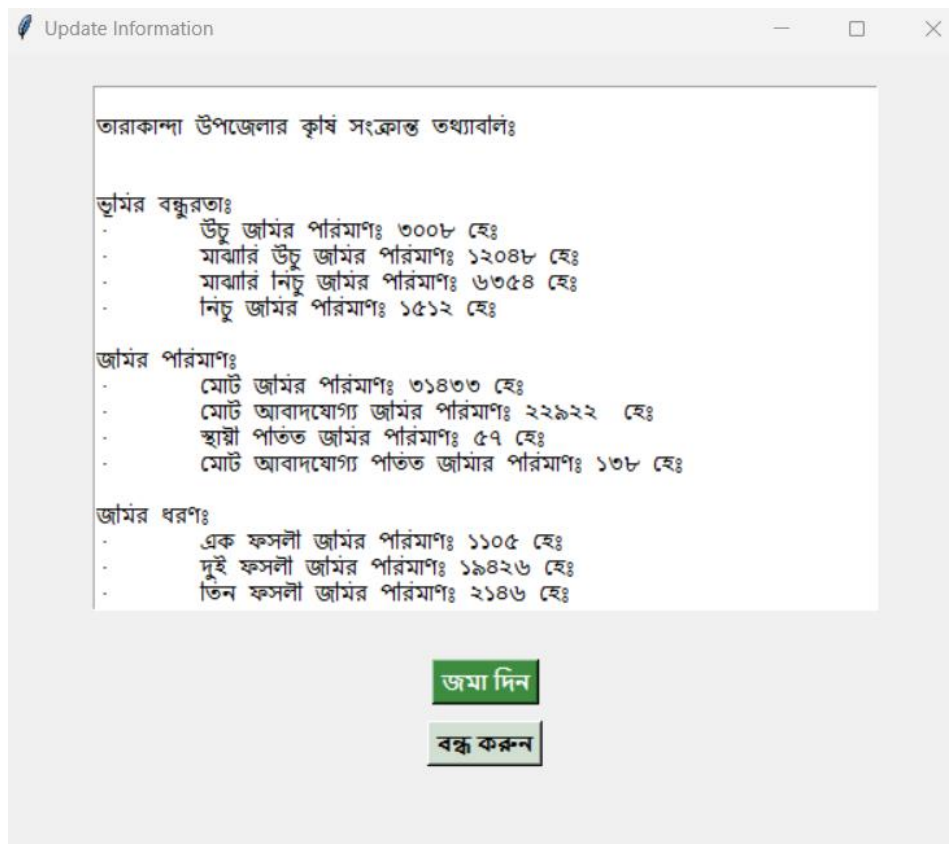
বন্ধ করুন

Admin Panel: In this section admin can update information by logging in with his username and password.



The image shows a login form for the Admin Panel. It is set against a background of a green field. The form has two input fields: 'এডমিন নাম:' (Admin Name) with the value 'admin' and 'পাসওয়ার্ড:' (Password) with masked characters '*****'. There are two buttons: 'লগ ইন' (Login) and 'Update Information'. Below the form, there is a label 'Admin Panel'.

Update Information Widget: In this widget admin can update the information as per his requirement.



The image shows a window titled 'Update Information'. It contains a list of agricultural data for 'তারাকান্দা উপজেলার কৃষি সংক্রান্ত তথ্যাবলিঃ' (Agriculture related information of Tarakananda Upazila). The data is organized into three sections: 'ভূমির বন্টনতাঃ' (Land Distribution), 'জমির পরিমাণঃ' (Land Area), and 'জমির ধরণঃ' (Land Type). Each section lists different types of land and their corresponding areas in hectares (হেঃ). At the bottom, there are two buttons: 'জমা দিন' (Submit) and 'বন্ধ করুন' (Close).

ভূমির বন্টনতাঃ	জমির পরিমাণঃ	জমির ধরণঃ
উঁচু জামির পরিমাণঃ ৩০০৮ হেঃ	মোট জামির পরিমাণঃ ৩১৪৩৩ হেঃ	এক ফসলী জামির পরিমাণঃ ১১০৫ হেঃ
মাঝারি উঁচু জামির পরিমাণঃ ১২০৪৮ হেঃ	মোট আবাদযোগ্য জামির পরিমাণঃ ২২৯২২ হেঃ	দুই ফসলী জামির পরিমাণঃ ১৯৪২৬ হেঃ
মাঝারি নিচু জামির পরিমাণঃ ৬৩৫৪ হেঃ	স্থায়ী পতিত জামির পরিমাণঃ ৫৭ হেঃ	তিন ফসলী জামির পরিমাণঃ ২১৪৬ হেঃ
নিচু জামির পরিমাণঃ ১৫১২ হেঃ	মোট আবাদযোগ্য পতিত জামির পরিমাণঃ ১৩৮ হেঃ	

Feedback Widget: In this widget, users can share their experiences, needs and opinions.



4.2 Development Tools and Environments

Adobe Photoshop: To convert image formats and resizing images.



Canva: For Background images.



Notepad: To write, store and update data.



Vs Code: IDE for code writing.



Chapter-5

CONCLUSION

5.1 Conclusion

The "SMART FARMING SYSTEM" project provides a user-friendly interface for accessing and updating agricultural information. With features like admin authentication, localized content, and communication tools, it addresses the specific needs of users in the Mymensingh district. The project lays a solid foundation for future enhancements and stands as a valuable tool for facilitating smart farming practices.

5.2 Limitations of this Project

- The GUI design is basic, and the styling options for buttons, labels, and windows are minimal
- The feedback mechanism is simplistic, storing all feedback in a single text file
- A major limitation of the application is that it has been developed only for Mymensingh district

5.3 Future Work

- To overcome our limitations
- Extend the project to support mobile devices by creating a responsive design or developing a dedicated mobile application
- Improve the visualization of data using charts, graphs, and other graphical elements

REFERENCES

- [1] M. Dawson., (2010). Python Programming for the Absolute Beginner. 3rd Edition.
- [2] Learning Python Tkinter: <https://www.geeksforgeeks.org/python-tkinter-tutorial/>
- [3] Learning Python Pillow: <https://www.geeksforgeeks.org/python-pillow-tutorial/>